

OPTIMIZATION GEOMETRY IN MULTI-TASK LLM POST-TRAINING

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ABSTRACT

Reinforcement learning with Verifiable Reward (RLVR) and on-policy distillation (OPD) both optimize LLMs on student-generated trajectories, yet their multi-task behavior is poorly understood. We present a cross-task gradient interference analysis to show that PPO is governed by reward-gradient geometry, GRPO is controlled by both reward-gradient geometry and intra-group reward variance. OPD is driven by teacher-student discrepancy. OPD contracts continuously as the student approaches its teachers, while GRPO maintains unit advantage energy on active groups. Across [MODEL/TASK SETUP], we find [MAIN RESULT 1], [MAIN RESULT 2], and [MAIN RESULT 3]. Together, these results support a view of RLVR as task-selective specialization and OPD as low-variance consolidation, and provide diagnostics for multi-task objective design, teacher selection, and stable truncated distillation.

The key distinction is how task-specific learning signals evolve during optimization. PPO is governed by reward-gradient geometry, GRPO retains finite normalized advantages within mixed-reward groups while such groups become increasingly rare, OPD continuously attenuates the teacher-student discrepancy.

Through Eq. equation 1, these differences directly affect the optimization trajectory: negative cross-task terms reduce per-task progress, while persistent conflict can induce gradient cancelation and slow finite-budget optimization. However, gradient interference is a local optimization quantity and does not by itself imply a universal ordering of final generalization performance among PPO, GRPO, and OPD.

RLVR and OPD induce fundamentally different gradient fields over the shared parameter space, leading to different interference dynamics during multi-task post-training.

1 GRADIENT INTERFERENCE IN MULTI-TASK LEARNING

Consider M tasks sampled with probabilities p_i . For task i , let $F_i(\theta)$ denote the objective to maximize and $g_i(\theta) = \nabla_{\theta} F_i(\theta)$ its gradient. The multi-task gradient is $g(\theta) = \sum_{j=1}^M p_j g_j(\theta)$. For a small update $\theta^+ = \theta + \eta g$, the improvement on task i is

$$F_i(\theta^+) - F_i(\theta) = \eta \left(p_i \|g_i\|_2^2 + \sum_{j \neq i} p_j \underbrace{\langle g_i, g_j \rangle}_{I_{ij}} \right) + O(\eta^2). \quad (1)$$

$I_{ij} < 0$ indicates gradient conflict between tasks i and j . In particular, negative aggregate cross-task interference $\sum_{j \neq i} p_j I_{ij}$ offsets the progress induced by task i 's own gradient. If $\sum_{j \neq i} p_j I_{ij} < -p_i \|g_i\|_2^2$, the multi-task update decreases F_i . We next characterize g_i and I_{ij} for PPO, GRPO, and OPD.

1.1 PPO

At the beginning of each PPO update (Schulman et al., 2017), $\theta = \theta_{\text{old}}$, so the importance ratio is one and clipping is locally inactive. Ignoring advantage-estimation error and optional KL regularization,

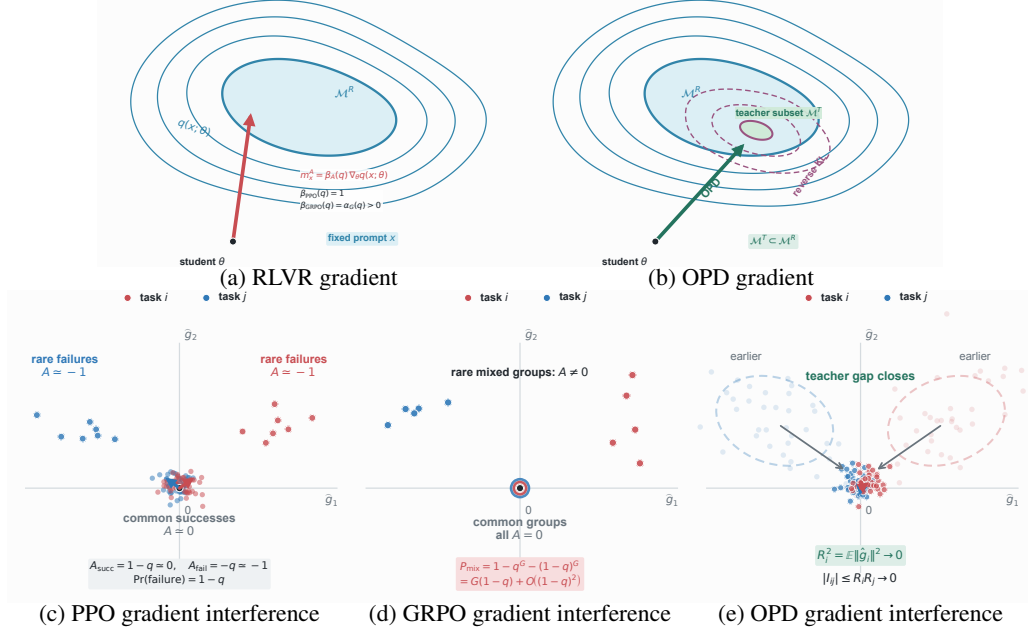


Figure 1: (a–b) RLVR and OPD are driven by different optimization vector fields. RLVR follows a reward-induced gradient field. OPD follows a teacher-induced gradient field. (c–e) scenarios where cross-task interference fades as learning progresses. In PPO, individual-response signal becomes rare. In GRPO, gradients arise only from increasingly rare groups containing both successes and failures. In OPD, the gradients contract to the origin as the student gets close to the teacher.

PPO reduces to the standard on-policy policy gradient,

$$g_i^{\text{PPO}} = \mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_\theta(\cdot|x)} [A_i(x, y) \nabla_\theta \log \pi_\theta(y | x)] = \nabla_\theta J_i(\theta), \quad (2)$$

where $J_i(\theta) = \mathbb{E}[r_i(x, y)]$.

For binary rewards, let $q_i(x) = \Pr[r_i(x, y) = 1]$ and use the exact per-prompt baseline, so that $A_i(x, y) = r_i(x, y) - q_i(x)$. Hence,

$$A_i(x, y) = \begin{cases} 1 - q_i(x), & r_i = 1, \\ -q_i(x), & r_i = 0. \end{cases}$$

As $q_i(x) \rightarrow 1$, successful rollouts have vanishing advantage, while failures retain $O(1)$ negative advantage but occur with probability $1 - q_i(x) \rightarrow 0$. Under a bounded-score assumption, this implies $\|\nabla_\theta q_i(x)\| \rightarrow 0$, and therefore

$$g_i^{\text{PPO}} = \mathbb{E}_{x \sim \mathcal{D}_i} [\nabla_\theta q_i(x)], \quad I_{ij}^{\text{PPO}} = \langle g_i^{\text{PPO}}, g_j^{\text{PPO}} \rangle \rightarrow 0 \quad (3)$$

as the corresponding tasks become solved.

1.2 GRPO

GRPO (Shao et al., 2024) samples G rollouts $y^{(1)}, \dots, y^{(G)}$ for each prompt and computes group-normalized advantages from their rewards. At the beginning of an on-policy update, $\theta = \theta_{\text{old}}$, so the importance ratios equal one and clipping is inactive. Ignoring KL regularization, the GRPO gradient is

$$g_i^{\text{GRPO}} = \mathbb{E} \left[\frac{1}{G} \sum_{k=1}^G \hat{A}_i^{(k)} \frac{1}{|y^{(k)}|} \sum_{t=1}^{|y^{(k)}|} \nabla_\theta \log \pi_\theta(y_t^{(k)} | x, y_{<t}^{(k)}) \right]. \quad (4)$$

We focus on the effect of group-relative reward normalization and ignore the $1/|y|$ term in the analysis. For binary rewards $r_i(x, y) \in \{0, 1\}$, define the success probability of prompt x as

$$q_i(x) = \Pr_{y \sim \pi_\theta(\cdot|x)} [r_i(x, y) = 1] = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} [r_i(x, y)]. \quad (5)$$

Then the expected GRPO gradient satisfies

$$\mathbb{E} \left[\frac{1}{G} \sum_{k=1}^G \hat{A}_i^{(k)} \nabla_{\theta} \log \pi_{\theta}(y^{(k)} | x) \middle| x \right] = \alpha_G(q_i(x)) \nabla_{\theta} q_i(x), \quad (6)$$

where $\alpha_G(q) > 0$ is determined by the group size and the current success probability. Its closed form and derivation are given in Appendix A.1. Therefore,

$$\begin{aligned} g_i^{\text{GRPO}} &= \mathbb{E}_{x \sim \mathcal{D}_i} [\alpha_G(q_i(x)) \nabla_{\theta} q_i(x)], \\ I_{ij}^{\text{GRPO}} &= \mathbb{E}_{x \sim \mathcal{D}_i, x' \sim \mathcal{D}_j} [\alpha_G(q_i(x)) \alpha_G(q_j(x')) \langle \nabla_{\theta} q_i(x), \nabla_{\theta} q_j(x') \rangle]. \end{aligned} \quad (7)$$

Consider a nearly solved but not perfectly solved prompt, with $q_i(x) = 1 - \delta < 1$. As shown in Appendix A.1, $\alpha_G(q_i(x)) = \sqrt{G-1} + O(\delta)$, so the GRPO coefficient does not vanish as $q_i(x) \rightarrow 1$. When a nonzero update occurs, it can still induce cross-task interference through $\langle \nabla_{\theta} q_i, \nabla_{\theta} q_j \rangle$.

Besides, a prompt has a nonzero advantage only if its G rollouts contain at least one successful rollout and at least one failed rollout. We refer to such a group as an *informative group*. Its probability is

$$P_{\text{mix}}(q) = 1 - q^G - (1 - q)^G = G(1 - q) + O((1 - q)^2). \quad (8)$$

As $q \rightarrow 1$ with $q < 1$, informative groups become increasingly rare. *Importantly, suppressing task interference in PPO/GRPO requires prompts to be reliably solved over repeated rollouts. This is a stringent condition since many prompts may produce occasional failures.*

1.3 ON-POLICY DISTILLATION

We focus on reverse-KL OPD, which is adopted in DeepSeek-V4, MiMo-V2-Flash, and Kimi K3 (DeepSeek-AI et al., 2026; Ma et al., 2026; Team et al., 2026). For task i , let $x \sim \mathcal{D}_i$ be a prompt and $y \sim \pi_{\theta}(\cdot | x)$ an on-policy rollout. At token position t , let $c_t = (x, y_{<t})$ denote the prefix. For simplicity, we treat the on-policy prefixes as fixed, as is standard in practical OPD implementations. We write the objective as

$$F_i^{\text{OPD}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t} [D_{\text{KL}}(\pi_{\theta}(\cdot | c_t) \| \pi_i^{\text{T}}(\cdot | c_t))]. \quad (9)$$

Using $\mathbb{E}_{v \sim \pi_{\theta}(\cdot | c_t)} [\nabla_{\theta} \log \pi_{\theta}(v | c_t)] = 0$, the task gradient can be written as

$$g_i^{\text{OPD}} = \mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t, v \sim \pi_{\theta}(\cdot | c_t)} \left[\log \frac{\pi_i^{\text{T}}(v | c_t)}{\pi_{\theta}(v | c_t)} \nabla_{\theta} \log \pi_{\theta}(v | c_t) \right]. \quad (10)$$

By Cauchy-Schwarz,

$$\|g_i^{\text{OPD}}\|_2 \leq \sqrt{\mathbb{E} \left[\log^2 \frac{\pi_i^{\text{T}}(v | c_t)}{\pi_{\theta}(v | c_t)} \right]} \sqrt{\mathbb{E} [\|\nabla_{\theta} \log \pi_{\theta}(v | c_t)\|_2^2]}. \quad (11)$$

If $\mathbb{E} \left[\log^2 \frac{\pi_i^{\text{T}}(v | c_t)}{\pi_{\theta}(v | c_t)} \right] \rightarrow 0$ and $\mathbb{E} [\|\nabla_{\theta} \log \pi_{\theta}(v | c_t)\|_2^2]$ is bounded, then $\|g_i^{\text{OPD}}\|_2 \rightarrow 0$ and $|I_{ij}^{\text{OPD}}| \leq \|g_i^{\text{OPD}}\|_2 \|g_j^{\text{OPD}}\|_2 \rightarrow 0$. As the teacher-student log-ratio vanishes, both the task gradient and its interference with other bounded task gradients vanish.

2 EXPERIMENTS

3 THE APPROACHABILITY OF NO-INTERFERENCE STATE

4 DESIGN PRINCIPLES

interference is not necessarily bad for end-to-end performance.

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A APPENDIX

A.1 DERIVATION OF THE GRPO GRADIENT UPDATE

We derive the expected GRPO gradient with binary reward used in Equation 6. Define

$$q(x; \theta) := \Pr_{y \sim \pi_\theta(\cdot | x)} [r(x, y) = 1]. \quad (12)$$

Let G responses be independently sampled from $\pi_\theta(\cdot | x)$, and let K denote the number of responses with reward 1. Conditional on $0 < K < G$, the group reward mean and standard deviation are

$$\bar{r} = \frac{K}{G}, \quad \sigma_r = \frac{\sqrt{K(G-K)}}{G}. \quad (13)$$

Therefore, the standardized advantages for successful and unsuccessful responses are respectively

$$A^+ = \frac{1 - \bar{r}}{\sigma_r} = \sqrt{\frac{G-K}{K}}, \quad A^- = \frac{-\bar{r}}{\sigma_r} = -\sqrt{\frac{K}{G-K}}. \quad (14)$$

The corresponding group update is

$$\hat{g}^{\text{GRPO}} = \frac{1}{G} \sum_{k=1}^G \hat{A}^{(k)} \nabla_\theta \log \pi_\theta(y^{(k)} | x). \quad (15)$$

We next relate the conditional score-function expectations to the gradient of the success probability. Since

$$q(x; \theta) = \sum_y \pi_\theta(y | x) \mathbf{1}\{r(x, y) = 1\}, \quad (16)$$

we have

$$\nabla_\theta q(x; \theta) = \sum_{y: r(x, y)=1} \pi_\theta(y | x) \nabla_\theta \log \pi_\theta(y | x). \quad (17)$$

Hence

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(y | x) | r = 1] = \frac{\nabla_\theta q(x; \theta)}{q(x; \theta)}. \quad (18)$$

Similarly,

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(y | x) | r = 0] = \frac{\nabla_\theta(1 - q(x; \theta))}{1 - q(x; \theta)} = -\frac{\nabla_\theta q(x; \theta)}{1 - q(x; \theta)}. \quad (19)$$

Conditioned on K , there are K successful responses and $G - K$ unsuccessful responses. Taking the expectation of Eq. equation 15 conditional on (x, K) and using Eqs. equation 14–equation 19 gives

$$\begin{aligned} \mathbb{E}[\hat{g}^{\text{GRPO}} | x, K] &= \frac{1}{G} \left[KA^+ \frac{\nabla_\theta q}{q} + (G - K)A^- \left(-\frac{\nabla_\theta q}{1 - q} \right) \right] \\ &= \frac{\sqrt{K(G - K)}}{G} \left(\frac{1}{q} + \frac{1}{1 - q} \right) \nabla_\theta q \\ &= \boxed{\frac{\sqrt{K(G - K)}}{Gq(1 - q)} \nabla_\theta q}. \end{aligned} \quad (20)$$

Here and below q denotes $q(x; \theta)$.

Finally,

$$K \sim \text{Binomial}(G, q). \quad (21)$$

Averaging Eq. equation 20 over K gives

$$\begin{aligned} \mathbb{E}[\hat{g}^{\text{GRPO}} | x] &= \frac{1}{Gq(1 - q)} \sum_{k=1}^{G-1} \binom{G}{k} q^k (1 - q)^{G-k} \sqrt{k(G - k)} \nabla_\theta q \\ &= \boxed{\alpha_G(q) \nabla_\theta q}, \end{aligned} \quad (22)$$

where

$$\alpha_G(q) := \frac{1}{Gq(1 - q)} \sum_{k=1}^{G-1} \binom{G}{k} q^k (1 - q)^{G-k} \sqrt{k(G - k)} > 0. \quad (23)$$

The cases $K = 0$ and $K = G$ do not contribute because all rewards in the group are identical and the centered group advantage is zero.

Therefore, for each prompt, the population GRPO update is a positive rescaling of the success-probability gradient:

$$\mathbb{E}[\hat{g}^{\text{GRPO}} | x] \parallel \nabla_\theta q(x; \theta). \quad (24)$$

For task i :

$$g_i^{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_i} [\alpha_G(q_i(x; \theta)) \nabla_\theta q_i(x; \theta)]. \quad (25)$$

GRPO preserves the reward-ascent direction for each prompt while changing its magnitude as a function of the current success probability.

A.2 GRADIENT INTERFERENCE OF OPD VARIANTS

We compare the cross-task gradient interference of full-vocabulary reverse-KL, sampled-token reverse-KL, and renormalized Top- K reverse-KL. Define

$$A_i(c_t, v) = \log \frac{\pi_i^T(v | c_t)}{\pi_\theta(v | c_t)}, \quad s_\theta(c_t, v) = \nabla_\theta \log \pi_\theta(v | c_t). \quad (26)$$

Full-vocabulary reverse KL. For a prefix c_t , the reverse-KL objective is

$$-D_{\text{KL}}(\pi_{\theta}(\cdot | c_t) \| \pi_i^{\text{T}}(\cdot | c_t)). \quad (27)$$

Its gradient is

$$\begin{aligned} & -\nabla_{\theta} D_{\text{KL}}(\pi_{\theta}(\cdot | c_t) \| \pi_i^{\text{T}}(\cdot | c_t)) \\ &= \sum_{v \in \mathcal{V}} \pi_{\theta}(v | c_t) A_i(c_t, v) s_{\theta}(c_t, v), \end{aligned} \quad (28)$$

where we use

$$\sum_{v \in \mathcal{V}} \pi_{\theta}(v | c_t) s_{\theta}(c_t, v) = 0. \quad (29)$$

Averaging the token-level gradient over the on-policy samples gives the task update direction:

$$g_i^{\text{full}} = \mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t, v \sim \pi_{\theta}(\cdot | c_t)} [A_i(c_t, v) s_{\theta}(c_t, v)]. \quad (30)$$

For tasks i and j , let $c_t = (x, y_{<t})$ and $c'_t = (x', y'_{<t'})$. Their gradient interference is

$$\begin{aligned} I_{ij}^{\text{full}} &= \langle g_i^{\text{full}}, g_j^{\text{full}} \rangle \\ &= \mathbb{E} [A_i(c_t, v) A_j(c'_t, u) \langle s_{\theta}(c_t, v), s_{\theta}(c'_t, u) \rangle], \end{aligned} \quad (31)$$

where the expectation is taken over independently drawn on-policy samples from tasks i and j .

By Jensen's inequality and Cauchy-Schwarz,

$$\begin{aligned} \|g_i^{\text{full}}\|_2 &\leq \mathbb{E} [A_i(c_t, v) \|s_{\theta}(c_t, v)\|_2] \\ &\leq \sqrt{\mathbb{E}[A_i(c_t, v)^2]} \sqrt{\mathbb{E}[\|s_{\theta}(c_t, v)\|_2^2]}. \end{aligned} \quad (32)$$

If $\mathbb{E}[A_i(c_t, v)^2] \rightarrow 0$ and the second moment of the score function remains bounded, then $\|g_i^{\text{full}}\|_2 \rightarrow 0$. So

$$|I_{ij}^{\text{full}}| \leq \|g_i^{\text{full}}\|_2 \|g_j^{\text{full}}\|_2 \rightarrow 0 \quad (33)$$

for any task j whose gradient remains bounded.

Sampled-token reverse-KL. Sampled-token reverse-KL draws a token $v \sim \pi_{\theta}(\cdot | c_t)$ and uses the teacher-student log-ratio as the token-level advantage:

$$\hat{g}_i^{\text{sample}} = A_i(c_t, v) s_{\theta}(c_t, v). \quad (34)$$

Taking expectation over the sampled token gives

$$\mathbb{E}_{v \sim \pi_{\theta}(\cdot | c_t)} [\hat{g}_i^{\text{sample}}] = \sum_{v \in \mathcal{V}} \pi_{\theta}(v | c_t) A_i(c_t, v) s_{\theta}(c_t, v). \quad (35)$$

So

$$g_i^{\text{sample}} = g_i^{\text{full}}, \quad (36)$$

and their gradient interference is identical:

$$I_{ij}^{\text{sample}} = I_{ij}^{\text{full}}. \quad (37)$$

The difference lies in stochasticity. The squared norm of a single-token update is

$$\|\hat{g}_i^{\text{sample}}\|_2^2 = A_i(c_t, v)^2 \|s_{\theta}(c_t, v)\|_2^2. \quad (38)$$

Thus sampled-token reverse-KL is an unbiased Monte Carlo estimator of the full-vocabulary direction, but individual updates may have higher sampling variance.

Renormalized Top- K reverse KL. Let $\mathcal{S}_{i,K}(c_t) \subseteq \mathcal{V}$ denote the Top- K vocabulary set. Define

$$Z_{\theta,i,K}(c_t) = \sum_{u \in \mathcal{S}_{i,K}(c_t)} \pi_{\theta}(u | c_t), \quad Z_{i,K}^T(c_t) = \sum_{u \in \mathcal{S}_{i,K}(c_t)} \pi_i^T(u | c_t), \quad (39)$$

and

$$\bar{\pi}_{\theta,K}(v | c_t) = \frac{\pi_{\theta}(v | c_t)}{Z_{\theta,i,K}(c_t)}, \quad \bar{\pi}_{i,K}^T(v | c_t) = \frac{\pi_i^T(v | c_t)}{Z_{i,K}^T(c_t)}. \quad (40)$$

The renormalized Top- K objective is

$$F_i^{\text{renK}} = -\mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t} [D_{\text{KL}}(\bar{\pi}_{\theta,K}(\cdot | c_t) \| \bar{\pi}_{i,K}^T(\cdot | c_t))]. \quad (41)$$

Its gradient can be written as

$$g_i^{\text{renK}} = \mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t, v \sim \bar{\pi}_{\theta,K}(\cdot | c_t)} [\bar{A}_{i,K}(c_t, v) \nabla_{\theta} \log \bar{\pi}_{\theta,K}(v | c_t)], \quad (42)$$

where

$$\bar{A}_{i,K}(c_t, v) = \log \frac{\bar{\pi}_{i,K}^T(v | c_t)}{\bar{\pi}_{\theta,K}(v | c_t)}. \quad (43)$$

The normalized score function satisfies

$$\nabla_{\theta} \log \bar{\pi}_{\theta,K}(v | c_t) = s_{\theta}(c_t, v) - \mathbb{E}_{u \sim \bar{\pi}_{\theta,K}(\cdot | c_t)} [s_{\theta}(c_t, u)]. \quad (44)$$

Moreover,

$$\bar{A}_{i,K}(c_t, v) = A_i(c_t, v) + \log \frac{Z_{\theta,i,K}(c_t)}{Z_{i,K}^T(c_t)}. \quad (45)$$

The second term is constant across tokens in the Top- K set and disappears after centering. So

$$g_i^{\text{renK}} = \mathbb{E}_{x \sim \mathcal{D}_i, y \sim \pi_{\theta}(\cdot | x), t, v \sim \bar{\pi}_{\theta,K}(\cdot | c_t)} [\tilde{A}_{i,K}(c_t, v) s_{\theta}(c_t, v)], \quad (46)$$

where

$$\tilde{A}_{i,K}(c_t, v) = A_i(c_t, v) - \mathbb{E}_{u \sim \bar{\pi}_{\theta,K}(\cdot | c_t)} [A_i(c_t, u)]. \quad (47)$$

Therefore, the corresponding cross-task interference is

$$\begin{aligned} I_{ij}^{\text{renK}} &= \langle g_i^{\text{renK}}, g_j^{\text{renK}} \rangle \\ &= \mathbb{E} [\tilde{A}_{i,K}(c_t, v) \tilde{A}_{j,K}(c'_t, u) \langle s_{\theta}(c_t, v), s_{\theta}(c'_t, u) \rangle]. \end{aligned} \quad (48)$$

By Cauchy–Schwarz,

$$\|g_i^{\text{renK}}\|_2 \leq \sqrt{\mathbb{E}[\tilde{A}_{i,K}(c_t, v)^2]} \sqrt{\mathbb{E}[\|s_{\theta}(c_t, v)\|_2^2]}. \quad (49)$$

As the normalized student distribution approaches the normalized teacher distribution on the Top- K set,

$$\bar{\pi}_{\theta,K}(\cdot | c_t) \rightarrow \bar{\pi}_{i,K}^T(\cdot | c_t), \quad (50)$$

we have

$$\tilde{A}_{i,K}(c_t, v) \rightarrow 0, \quad \|g_i^{\text{renK}}\|_2 \rightarrow 0, \quad (51)$$

and

$$I_{ij}^{\text{renK}} \rightarrow 0 \quad (52)$$

for any task j whose gradient remains bounded.

Unlike full-vocabulary OPD, this condition only requires the relative probabilities in the Top- K set to match. $\bar{\pi}_{\theta,K} = \bar{\pi}_{i,K}^T$ is equivalent to

$$\frac{\pi_i^T(v | c_t)}{\pi_{\theta}(v | c_t)} = \text{constant}, \quad v \in \mathcal{S}_{i,K}(c_t). \quad (53)$$

Renormalized Top- K reverse-KL can have a vanishing gradient even when the student and teacher still differ in the total vocabulary distribution.

A.3 RL TEACHER AND OPD EQUIVALENCE

We consider an idealized setting where an RL teacher is obtained from the same policy used to initialize the OPD student. Let π_0 denote this shared initialization and the RL reference policy. Consider the KL-regularized RL objective

$$J_{\text{RL}}(\pi) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x)} [r(x, y)] - \beta \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi(\cdot | x) \| \pi_0(\cdot | x))], \quad (54)$$

where $\beta > 0$ is the KL penalty coefficient. Assume that the teacher π^T is the exact optimum of Eq. equation 54. For each prompt x , the optimal policy takes the standard Boltzmann form

$$\pi^T(y | x) = \frac{\pi_0(y | x) \exp(r(x, y)/\beta)}{Z(x)}, \quad Z(x) = \mathbb{E}_{y \sim \pi_0(\cdot | x)} [e^{r(x, y)/\beta}]. \quad (55)$$

Now consider reverse-KL OPD toward this fixed teacher,

$$F_{\text{OPD}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi^T(\cdot | x))]. \quad (56)$$

Substituting Eq. equation 55 gives

$$\begin{aligned} F_{\text{OPD}}(\theta) &= \mathbb{E}_{x, y \sim \pi_\theta} \left[\frac{r(x, y)}{\beta} - \log \frac{\pi_\theta(y | x)}{\pi_0(y | x)} - \log Z(x) \right] \\ &= \frac{1}{\beta} J_{\text{RL}}(\pi_\theta) - \mathbb{E}_{x \sim \mathcal{D}} [\log Z(x)]. \end{aligned} \quad (57)$$

Since $Z(x)$ is independent of θ ,

$$g_{\text{OPD}}(\theta) = \frac{1}{\beta} g_{\text{RL}}(\theta). \quad (58)$$

Thus, under an exact KL-regularized RL teacher, reverse-KL OPD recovers the same policy gradient as KL-regularized RL up to a constant scale.

Sampled-token variance. Let $s_\theta(x, y) = \nabla_\theta \log \pi_\theta(y | x)$ and define the KL-regularized reward

$$\tilde{r}_\theta(x, y) = r(x, y) - \beta \log \frac{\pi_\theta(y | x)}{\pi_0(y | x)}. \quad (59)$$

The corresponding policy gradient can be written as

$$g_{\text{RL}} = \mathbb{E}[(\tilde{r}_\theta(x, y) - b(x)) s_\theta(x, y)], \quad (60)$$

for any prompt-dependent baseline $b(x)$, since $\mathbb{E}_{y \sim \pi_\theta} [s_\theta(x, y)] = 0$.

From Eq. equation 55,

$$\log \frac{\pi^T(y | x)}{\pi_\theta(y | x)} = \frac{1}{\beta} (\tilde{r}_\theta(x, y) - \beta \log Z(x)). \quad (61)$$

Therefore, choosing the matched baseline $b(x) = \beta \log Z(x)$ gives

$$\hat{g}_{\text{OPD}} = \frac{1}{\beta} \hat{g}_{\text{RL}}, \quad (62)$$

and hence

$$\text{Cov}[\hat{g}_{\text{OPD}} | x] = \frac{1}{\beta^2} \text{Cov}[\hat{g}_{\text{RL}} | x]. \quad (63)$$

Importantly, Eq. equation 63 does not imply an intrinsic variance advantage for either method. The mean gradient is scaled by the same factor, so

$$\frac{\text{tr Cov}[\hat{g}_{\text{OPD}} | x]}{\|\hat{g}_{\text{OPD}}(x)\|_2^2} = \frac{\text{tr Cov}[\hat{g}_{\text{RL}} | x]}{\|\hat{g}_{\text{RL}}(x)\|_2^2}. \quad (64)$$

Equivalently, the factor $1/\beta$ can be absorbed into the learning rate. Thus, the absolute variance scaling in Eq. equation 63 is simply a consequence of objective scaling and does not by itself imply better or worse stochastic optimization.

Full-vocabulary OPD reduces token-sampling variance. The distinction changes when OPD uses the full teacher distribution rather than a sampled target token. Consider an on-policy prefix $c = (x, y_{<t})$, and define

$$A_{\text{OPD}}(c, v) = \log \frac{\pi^T(v | c)}{\pi_\theta(v | c)} - \mathbb{E}_{v' \sim \pi_\theta(\cdot | c)} \left[\log \frac{\pi^T(v' | c)}{\pi_\theta(v' | c)} \right]. \quad (65)$$

The centering term is a prefix-dependent baseline and therefore does not change the expected gradient. Let

$$h(c, v) = A_{\text{OPD}}(c, v) \nabla_\theta \log \pi_\theta(v | c). \quad (66)$$

A sampled-token OPD estimator draws $V \sim \pi_\theta(\cdot | c)$ and uses

$$\hat{g}_{\text{OPD}}^{\text{sample}}(c) = h(c, V), \quad (67)$$

whereas full-vocabulary reverse KL computes

$$g_{\text{OPD}}^{\text{full}}(c) = \sum_{v \in \mathcal{V}} \pi_\theta(v | c) h(c, v) = \mathbb{E} \left[\hat{g}_{\text{OPD}}^{\text{sample}}(c) | c \right]. \quad (68)$$

Thus, full-vocabulary OPD is the conditional expectation of the sampled-token estimator given the same prefix. By the law of total covariance,

$$\begin{aligned} \text{Cov} \left[\hat{g}_{\text{OPD}}^{\text{sample}} \right] &= \text{Cov} \left[g_{\text{OPD}}^{\text{full}}(c) \right] \\ &\quad + \mathbb{E}_c \left[\text{Cov} \left[\hat{g}_{\text{OPD}}^{\text{sample}} | c \right] \right], \end{aligned} \quad (69)$$

which immediately gives

$$\text{Cov} \left[g_{\text{OPD}}^{\text{full}}(c) \right] \preceq \text{Cov} \left[\hat{g}_{\text{OPD}}^{\text{sample}} \right]. \quad (70)$$